

Networking Task 01: Understanding Your Network Environment

Objective

The purpose of this task is to understand the basic components of a network, identify the network configuration of my system, and perform basic network connectivity tests.

Part A: Network Information

System Network Details

Detail	Value
Hostname (Device Name)	E3
IPv4 Address	172.21.11.65
MAC Address	08-8F-C3-B5-7E-CC
Default Gateway	172.21.11.1
DNS Server	172.21.3.100

Part B: Basic Networking Concepts

What is an IP Address?

An IP (Internet Protocol) Address is a unique numerical identifier assigned to a device connected to a network. It allows devices to communicate and exchange data over a network or the internet.

What is a MAC Address?

A MAC (Media Access Control) Address is a unique hardware address assigned to a network interface card by the manufacturer. It is used to identify devices on a local network.

What is a Default Gateway?

A Default Gateway is a networking device, usually a router, that connects a local network to other networks such as the internet. It acts as the exit point for traffic leaving the local network.

What is DNS?

DNS (Domain Name System) is a service that translates human-readable domain names such as google.com into IP addresses that computers use to locate servers on a network.

Difference Between Public IP and Private IP

Public IP	Private IP
Assigned by an Internet Service Provider (ISP)	Assigned within a local network
Accessible from the internet	Used only within a private network
Must be globally unique	Can be reused across different private networks
Used to identify a network on the internet	Used for communication inside a local network

Part C: Basic Network Diagram

Network Topology

Internet
↓
Router / Gateway (172.21.11.1)
↓
Device (E3)
IP Address: 172.21.11.6

Part D: Network Connectivity Test

Commands Used

```
C:\Users\KJC Staff>tracert google.com

Tracing route to google.com [142.251.42.238]
over a maximum of 30 hops:

  0  *          *          *          Request timed out.
  1  <1 ms     <1 ms     <1 ms     172.21.3.100
  2  <1 ms     <1 ms     <1 ms     103.5.132.177
  3  <1 ms     <1 ms     <1 ms     124.40.245.250
  4  15 ms     15 ms     17 ms     209.85.149.210
  5  14 ms     14 ms     15 ms     142.251.76.23
  6  15 ms     14 ms     14 ms     142.250.214.109
  7  *          13 ms     13 ms     pnbomb-aw-in-f14.1e100.net [142.251.42.238]
  8

Trace complete.
```

```
C:\Users\KJC Staff>tracert google.com
```

```
Tracing route to google.com [142.251.42.238]  
over a maximum of 30 hops:
```

```
 1  *      *      *      Request timed out.  
 2  <1 ms   <1 ms   <1 ms   172.21.3.100  
 3  <1 ms   <1 ms   <1 ms   103.5.132.177  
 4  <1 ms   <1 ms   <1 ms   124.40.245.250  
 5  15 ms   15 ms   17 ms   209.85.149.210  
 6  14 ms   14 ms   15 ms   142.251.76.23  
 7  15 ms   14 ms   14 ms   142.250.214.109  
 8  *      13 ms   13 ms   pnbomb-aw-in-f14.1e100.net [142.251.42.238]
```

```
Trace complete.
```

```
C:\Users\KJC Staff>ipconfig/all
```

```
Windows IP Configuration
```

```
Host Name . . . . . : E3  
Primary Dns Suffix . . . . . :  
Node Type . . . . . : Hybrid  
IP Routing Enabled. . . . . : No  
WINS Proxy Enabled. . . . . : No
```

```
Ethernet adapter Ethernet:
```

```
Connection-specific DNS Suffix . . :  
Description . . . . . : Realtek PCIe GbE Family Controller  
Physical Address. . . . . : 08-8F-C3-B5-7E-CC  
DHCP Enabled. . . . . : No  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::6036:6757:c63b:3fde%8(Preferred)  
IPv4 Address. . . . . : 172.21.11.65(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 172.21.11.1  
DHCPv6 IAID . . . . . : 168333251  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-48-FB-AC-08-8F-C3-B5-7E-CC  
DNS Servers . . . . . : 172.21.3.100  
NetBIOS over Tcpip. . . . . : Enabled
```

```
Ethernet adapter Ethernet 2:
```

```
Connection-specific DNS Suffix . . :  
Description . . . . . : VirtualBox Host-Only Ethernet Adapter  
Physical Address. . . . . : 0A-00-27-00-00-10  
DHCP Enabled. . . . . : No  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::4945:ecea:877:b857%16(Preferred)  
IPv4 Address. . . . . : 192.168.56.1(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
DHCPv6 IAID . . . . . : 336199719  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-48-FB-AC-08-8F-C3-B5-7E-CC  
NetBIOS over Tcpip. . . . . : Enabled
```

```
C:\Users\KJC Staff>ping google.com
```

```
Pinging google.com [142.251.42.238] with 32 bytes of data:
```

```
Reply from 142.251.42.238: bytes=32 time=13ms TTL=118
```

```
Reply from 142.251.42.238: bytes=32 time=14ms TTL=118
```

```
Reply from 142.251.42.238: bytes=32 time=13ms TTL=118
```

```
Reply from 142.251.42.238: bytes=32 time=14ms TTL=118
```

```
Ping statistics for 142.251.42.238:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:
```

```
    Minimum = 13ms, Maximum = 14ms, Average = 13ms
```

Connectivity Test Results

1. Was the ping successful?

Yes. The ping was successful. All packets were transmitted and received successfully without packet loss.

2. How many hops were shown?

30

3. What is the purpose of traceroute?

Traceroute is a network diagnostic tool used to trace the path that data packets take from a source device to a destination. It helps identify routing paths, delays, and network connectivity issues.

Conclusion

This task helped me understand the basic concepts of computer networking, identify my system's network configuration, and perform network connectivity tests using ping and traceroute. I learned how devices communicate within a network and how data travels across different network paths to reach its destination.